MEMORANDUM OF AGREEMENT

Project Title: "In vitro and In Vivo Anti-hypercholesterolemic activity of *Alocasia macrorrhiza* (Araceae) and its Food Products"

KNOW ALL MEN BY THESE PRESENTS:

This **Memorandum of Agreement** is made and entered into and executed this _____ day of _____, at _____ by and among:

The **PHILIPPINE COUNCIL FOR HEALTH RESEARCH AND DEVELOPMENT,** one of the sectoral councils of the **DEPARTMENT OF SCIENCE & TECHNOLOGY (DOST)** hereinafter referred to as **PCHRD-DOST,** represented by its Executive Director. **DR. JAIME C. MONTOYA,** with postal address at 3/F DOST Building, Bicutan, Taguig City

The **DEPARTMENT OF SCIENCE AND TECHNOLOGY REGIONAL OFFICE NO. V**, on behalf of the Bicol Consortium for Health Research and Development, Region V (RHRDC-V), hereinafter referred to as **DOST-V** represented by its Regional Director, **MR. ROMMEL R. SERRANO**, with postal address at Regional Center Site, Rawis, Legazpi City;

and

The **PARTIDO STATE UNIVERSITY (PSU)**, herein referred to as the **PROPONENT**, represented by its **President, RAUL G. BRADECINA, PhD** with postal address at San Juan Bautista, Goa, Camarines Sur;

with

DIOMERL EDWARD BALDO herein referred to as **PROJECT LEADER**, with postal address at Partido State University, San Juan Bautista, Goa, Camarines Sur;

WITNESSETH

WHEREAS, **PCHRD-DOST** is committed to develop and enhance the research capacity of institutions in the regions as a strategy to improve the health conditions of Filipinos;

WHEREAS, **DOST V** provides the regional presence of the DOST in Region V and is the focal point and financial administrator of the health research committee in the region;

WHEREAS, the **PROPONENT** has submitted a proposal to BCHRD, entitled "***In vitro* and *In Vivo* Anti-hypercholesterolemic activity of *Alocasia macrorrhiza* (Araceae) and its Food Products**" through PCHRD's Regional Research Fund (RRF) mechanism;

WHEREAS, the **BCHRD** reviewed the technical and ethical merits of the above-titled **PROJECT**, and endorsed the same to **DOST-PCHRD**;

WHEREAS, the **PROPONENT** has the staff and expertise to undertake the project and based on these, **BCHRD** and **PCHRD-DOST** approved the project;

WHEREAS, **PCHRD-DOST** has the necessary funds to support the project;

NOW, **THEREFORE**, for and in consideration of the foregoing premises, among parties hereto mutually agree to the following terms and conditions:

I. OBJECTIVES OF THE PROJECT:

The study generally aims to evaluate the anti-hypercholesterolemic activity of the crude extracts from the corms of *Alocasia macrorrhiza*. Specifically, it aims to:

1. Determine the phytochemical constituents of the *A. macrorrhiza* corm extracts;
2. Investigate the HMG-CoA reductase percentage of the plant extract using optimized enzymatic kits In vitro;
3. Assess the effects of *A. macrorrhiza* extracts and its food products on blood glucose, total cholesterol, triglycerides, and high-density lipoprotein (HDL), low density lipoprotein (LDL) and very low-density lipoprotein (VLDL).
4. Investigate the histopathological characteristics of the liver treated with different concentrations of *A. macrorrhiza* extract and its food products.

II. OPERATION OF THE PROJECT

The **PROJECT** shall be undertaken by the **PROPONENT**, through the Project Leader **DIOMERL EDWARD BALDO** in accordance with the approved **PROJECT** proposal hereto attached as **Annex "1"** and made an integral part hereof.

III. OBLIGATIONS OF PCHRD-DOST

1. Allocate the amount of **SIX HUNDRED FIFTY-SIX THOUSAND NINE HUNDRED FORTY-EIGHT PESOS and 84/100 (PhP656,948.84)** chargeable against the **PCHRD-DOST** Grants-in-Aid (PCHRD-DOST-GIA), in conformity with **DOST** Administrative Order (A.O.) No. 009, Series of 2017 hereto attached as **Annex "2"** and made an integral part hereof. This amount shall be expended as allocated in the approved Project Line-item Budget hereto attached as **Annex "3"** and made an integral part hereof. The amount covers the full cost of the project for personal services, maintenance and other operating expenses subject to existing government accounting and auditing laws, rules, and regulations. The amount shall be paid in accordance with Item VI of this Agreement.
2. Monitor the **PROJECT** activities and provide technical assistance relevant to the achievement of the **PROJECT** objectives; and
3. Provide other forms of assistance to the **PROJECT** in coordination with other government agencies in matters requiring their attention or cooperation insofar as these are relevant to the **PROJECT** and within the capability of **PCHRD-DOST**.

IV. OBLIGATIONS OF THE PROPONENT

1. Submit quarterly progress (using RRF Report form) and financial reports in three (3) hard copies and one (1) electronic copy to **DOST RO V** and **PCHRD-DOST** within one (1) month after each quarter of the implementation period;
2. Submit not later than 30 days after **PROJECT**'s completion to **DOST RO V** and **PCHRD-DOST** three (3) hard copies and one (1) electronic copy of technical report and a paper in publishable form;
3. Submit to **DOST RO V** not later than 30 days after PROJECT's completion, an Audited Financial Report;
4. Hire project personnel on contractual basis and co – terminus with this Agreement;
5. Ensure the continuity of the **PROJECT** and notify **DOST RO V** and **PCHRD-DOST** of instances such as where the senior project Personnel, including the Project Leader, will be absent from his/her station for a period exceeding ten (10) consecutive working days and designate a responsible project official to oversee the operations of the project in his/her absence;
6. Identify, protect, and manage all Intellectual Property (IP) generated and where applicable, pursue commercialization of R&D results in accordance with Technology Transfer Act of 2009 (Republic Act No.10055) and its IRR, other applicable laws; and
7. Notify **DOST RO V** and **PCHRD-DOST** within a reasonable time of all IPR application/s, licenses, and assignments made on the PROJECT results.

8. Deposit the grant provided by **DOST-PCHRD** to a government depository bank such as the Land Bank of the Philippines (LBP) or Development Bank of the Philippines (DBP); and
9. Ensure that the RRF approved projects are registered in the Health Research and Development Network (HERDIN Plus) at http://herdin.ph upon receipt of Notice of Approval; and update project information as necessary.

V. OBLIGATIONS OF DOST RO V and BCHRD

The **BCHRD** thru the **DOST RO V** shall:

1. Release the funds to the proponent based on agreed cost & terms of payment as specified in Item VI of this agreement;
2. Provide technical and administrative support to the **PROJECT**, in cooperation with the relevant sub-committees of **BCHRD**;
3. Monitor and review the technical and financial status of the **PROJECT** in accordance to the terms set forth in Item IX;
4. Submit to **PCHRD-DOST** the required final technical and audited financial reports a week after receipt of submitted documents by the **PROPONENT**;
5. Assist the **PROPONENT** to disseminate and translate research results to policies and /or actions.
6. Assist the **PROPONENT** to register the RRF approved projects in the Health Research and Development Network (HERDIN Plus) at http://herdin.ph; and update project information as necessary.

VI. COST AND TERMS OF PAYMENT

The **DOST-PCHRD** assistance to this Agreement is ***SIX HUNDRED FIFTY-SIX THOUSAND NINE HUNDRED FORTY-EIGHT PESOS and 84/100*** **(PhP656,948.84).** This amount shall be released by **PCHRD- DOST** to **DOST RO V.**

The **DOST RO V** shall administer the release of the **PROJECT** funds to the **PROPONENT** upon execution of this Memorandum of Agreement.

ACTIVITIES	EXPECTED OUTPUT	AMOUNT
Mobilization Fund	1. Signing of the Contract	80%
	2. Submission of Progress Technical and Financial Reports	
2nd Release	1. Report on analysis of data 2.Submission of final technical report, audited financial report certified correct by the accountant, and a publishable paper	20%

The Project Leader shall present the final technical report in an evaluation meeting to be organized by the **DOST-PCHRD** and **BCHRD**. The last tranche of funds will be released upon acceptance of a satisfactory final technical report.

VII. FUND UTILIZATION

Disbursement of grants shall be in accordance with the approved line – item budget (LIB) and subject to existing government accounting and auditing laws, rules and procedures *(DOST AO 009, Series of 2017).*

Request for budget reprogramming shall be governed by *No. 6.3.2 of DOST AO 009, Series of 2017.* It shall be made at least two (2) months before the end of the project period and shall not be done more than three (3) times in each implementation year (including approved extension) except for meritorious cases as approved by **PCHRD-DOST**. A revised line–item budget (LIB) shall be approved to cover any of the following: budget realignment/transfer of funds, and creation of expense item/s.

VIII. OWNERSHIP AND UTILIZATION OF PROJECT RESULT/ INTELLECTUAL PROPERTY RIGHTS (IPR) PROTECTION

The ownership of IPR and sharing of revenues of the **PROJECT** (if any) shall be vested with the **PROPONENT** and its researchers in accordance with Republic Act No. 10055, otherwise known as the *"Philippines Technology Transfer Act of 2009"*, hereto attached as **Annex "4"** and made an integral part hereof.

IX. PROJECT MONITORING

DOST RO V with the relevant committees of **BCHRD**, and **PCHRD–DOST** shall conduct periodic field visits and review technical and financial reports submitted by the **PROPONENT** to see to it that everything is in accordance with Item VI of this Agreement, and to resolve problems encountered (if any) in the **PROJECT** implementation.

The Commission on Audit shall exercise visitorial authority pursuant to Section 29 of Presidential Decree No. 1445 insofar as **PROJECT** outputs/performances under this Agreement are concerned.

X. PUBLICATION/PRESENTATION OF RESEARCH RESULTS

The **PROPONENT** is allowed to publish/present/make known to others the results/outputs of the research undertaken under this Agreement provided that the same does not contribute a prejudicial disclosure nor include the disclosure of confidential information, and shall identify the **PROPONENT**, **BCHRD**, and **PCHRD-DOST** as the sources of the output.

The **PROPONENT**, **DOST V**, **BCHRD,** and **PCHRD-DOST** reserve the right to use all the data and findings of the PROJECT in the pursuit if their respective academic/official functions.

XI. RECOGNITION OF PRINCIPAL AUTHORS

All major reports of the proponent shall be made in the name of the **PROPONENT** through its researchers/project leaders/authors.

Proper acknowledgment of the financial and other assistance by the **DOST RO V**, **BCHRD**, and **PCHRD- DOST** shall be given due credit in all reports, articles, and news release.

Ownership of copyright shall conform to Rule 6 of the Implementing Rules and Regulations (IRR) of Technology Transfer Act 2009 (RA 10055).

XII. PROJECT DURATION

The **PROJECT** shall be completed within the period of twelve (12) months covering the period **01 April 2020 to 31 March 2021**.

XIII. DISCONTINUANCE OF THE PROJECT

The **PCHRD-DOST** reserves the right to recommend discontinuance of the project implementation at anytime for the violation of this agreement or when it is determined that the results that are obtained do not justify further activity. Further, **PCHRD-DOST** has the authority to terminate the **PROJECT** when funds are not available from the national government.

XIV. PROJECT DELAYS

Except for delays resulting from circumstances beyond the control of the Parties, such as, but not limited to, force majeure and others of similar nature, no extension of the **PROJECT** period shall be allowed. Where such uncontrollable situations, however, arise, extension may be allowed at no additional **PROJECT** cost.

XV. THIRD PARTY PARTICIPATION

Any activity in the course of the implementation of the **PROJECT** assigned to or contributed by the third parties including but not limited to provision, import, export, transfer, design or use of equipment/softwares, collaborative research, or use of facilities should be disclosed by the **PROPONENT** to **PCHRD-DOST** prior to engaging in said activity. Any agreement entered into by the **PROPONENT** without prior consent of **BCHRD**, and **PCHRD-DOST** will not be binding on the latter insofar as confidentiality, treatment, ownership, enforcement and disposition of intellectual property rights resulting from the said activity is concerned. **PCHRD-DOST** reserves the right to withhold consent to such agreement(s) upon determination that it is not to the best interest of the herein Parties, or gives undue advantage to said third parties and for similar causes.

XVII. ALTERNATIVE MODES OF DISPUTE RESOLUTION AND VENUE

Any controversy between the **PROPONENT**, **DOST RO V, BCHRD, PSU,** and **PCHRD-DOST** arising from the interpretation and application of this agreement shall be settled in accordance to Republic Act No. 9285, also known as "Alternative Dispute Resolution Act of 2004" and its Implementing Rules and Regulation (IRR).

XVIII. EFFECTIVITY

This Agreement shall take effect on _____and shall be in full force and effect until the completion of the **PROJECT**.

IN WITNESS WHEREOF, the Parties have hereto set their hands on the date and at the place first above mentioned.

PHILIPPINE COUNCIL FOR HEALTH RESEARCH AND DEVELOPMENT

By:

JAIME C. MONTOYA, MD, MSc, PhD, CESO II
Executive Director

DEPARTMENT OF SCIENCE AND TECHNOLOGY RO V

By:

ROMMEL R. SERRANO
Regional Director

PARTIDO STATE UNIVERSITY

By:

RAUL G. BRADECINA
President

DIOMERL EDWARD BALDO
Project Leader

Signed in the Presence of:

CARINA L. REBULANAN
Chief, IDD-DOST-PCHRD

ARNULFO M. MASCARIÑAS
Consortium Chair, BCHRD

CERTIFIED FUNDS AVAILABLE:

JESSAMYN M. BUCLATIN

№20-0404

Title: In vitro and In-Vivo Anti-hypercholesterolemic Activity of *Alocasia macrorrhiza* (Araceae) and its Food Products

Proponent: Diomerl Edward Baldo

Duration: 0ne Year

LINE-ITEM BUDGET

Item	PCHRD Grant	PSU Counterpart
I. **Personal Services (PS)**		
Salaries		
1 Project Assistant I @20,109.60 x 12 months	241,315.20	
Honoraria		
1 Project leader @8,800/month x 12 months	105,600.00	
3 Project Staff Level 1 @4,800/month x 3 months	43,200.00	
Sub-total PS	**390,115.20**	
II. Maintenance and Other Operating Expenses (MOOE)		
Traveling Expense	30,000.00	
Supplies and Materials Expense	150,000.00	50,000.00
Communications Expense	5,000.00	
Representation Expense		10,000.00
Printing and Binding Expense		10,000.00
Other Professional Expense		
Analytical Services	30,000.00	
1 Statistician @3,000/consultation x 2 consultations	6,000.00	
Sub-total MOOE	**221,000.00**	
III. Indirect Cost (IC)		
Honoraria	**30,000.00**	
Utilities	**15,833.64**	
Sub-total IC	**45,833.64**	
GRAND TOTAL	**656,948.84**	**70,000.00**

Certified Funds Available:

02/20

JESSAMYN M. BUCLATIN
Accountant III

Recommended by:

CARINA L. REBULANAN
Chief, IDD

Approved:

JAIME C. MONTOYA, M.D., MSc, PhD, CESO III
Executive Director

ACKNOWLEDGMENT

Republic of the Philippines)
~~TAGUIG CITY~~) S.S.

BEFORE ME, a Notary Public in and for TAGUIG CITY, Philippines, personally appeared the following persons presenting to me competent proof of their identity:

Name	Evidence of Identity (ID)	Date/Place Issued
JAIME C. MONTOYA	~~GSIS ID No. CRN-006-0070-9230-3~~	~~Pasay City~~
ROMMEL R. SERRANO	Passport EC6518477	Legazpi City
RAUL G. BRADECINA	Passport P1359480A	Legazpi City
DIOMERL EDWARD BALDO		

all known to me and to me known to be the same persons who executed the foregoing instrument and they acknowledged to me that the same is their free and voluntary act and deed and that of the entities they respectively represent.

This instrument refers to a Memorandum of Agreement (MOA) consisting of six (6) pages including the page on which this Acknowledgment is written has been signed by the Parties and their instrumental witnesses on each and every page.

WITNESS MY HAND and SEAL this _____ day of 02 SEP 2020 2020, at the place first above written.

RACHEL GINA W. COPANUT-PANGWI
NOTARY PUBLIC UNTIL DEC. 31, 2021
Notarial Commission Appt. No. 10 (2020-2021)
2/F, Pacura Bldg. #427 Quezon Ave., Lower Bicutan, T.C.
PTR No. A-4689597/1-3-20, Taguig City
IBP O.R. No. 093183/10-14-19, RSM
MCLE Comp. No. VI-0019686/3-1-2019; ROLL No. 61627

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